

Case Ace v2.0 — Phase 10: Audio Redaction and Verification

1. Overview and Invariants

Phase 10 provides client-side acoustic redaction and mandatory verification of consultation audio before any speech data is transmitted to upstream processing services (such as Cloud Speech-to-Text v2).

It implements the following core constitutional invariants:

Constraint C1 / C4 (Volatile Memory Discipline & Immediate Release): Raw, unredacted consultation audio is stored strictly in memory and is completely zeroed and dereferenced once the redacted version is verified.

Constraint C8 (Fail-Closed Acoustic Verification): Audio redaction is verified by re-running local Pass 1 ASR on the redacted audio. If any identifier survives in the verification transcript, upload is strictly blocked, the session returns to the Phase 9 Redaction Review Gate with survivors highlighted, and the adviser must re-approve.

Configurable Padding ($\geq 250\text{ms}$ Enforcement): Word-level timestamps drift and acoustic transients precede and succeed phonetic boundaries. A minimum padding of 250ms is strictly enforced on both sides of every redaction span.

Interval Merging: Adjacent and overlapping redaction spans within padding distance are merged into contiguous silencing blocks to eliminate micro-gaps that could leak partial phonemes.

Region-Level Acoustic Assertions: Muted regions are mathematically verified in memory to ensure $\text{RMS} < 0.00001$ (pure digital silence) and exact duration preservation.

2. Architecture & Pipeline

flowchart TD

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A[Phase 9 Approved Spans & Timestamps] --> B[Audio Redaction Engine]
B --> C[Apply Padding >= 250ms & Merge Overlaps]
C --> D[Zero Audio Samples / Sine Bleed]
D --> E[Region Acoustic Energy Assertions]
E --> F[Verification Pass: Local Pass 1 ASR on Redacted Audio]
F --> G[Phase 8 Identifier Scanner on Verification Transcript]
G --> H{Survivors Detected?}
H -- Yes (> 0) --> I[FAIL CLOSED: Block Upload, Lock Gate, Flag Survivors]
I --> J[Return Adviser to Phase 9 Gate]
H -- No (0) --> K[Encode LINEAR16 WAV 16kHz]
K --> L[VolatileSessionStore: Zero & Release Raw Audio]
L --> M[Transition to Stage: 'audio_redacted']
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3. Mathematical & Acoustic Specifications

3.1 Padding Calculation & Interval Merging

For any approved redaction span i with audio bounds $[t_{\text{start}}, t_{\text{end}}]$:

$$t_{\text{pad}} = \max(250\text{ms}, t_{\text{config_padding}})$$

$$t'_{\text{start}} = \max(0.0, t_{\text{start}} - t_{\text{pad}})$$

$$t'_{\text{end}} = \min(t_{\text{duration}}, t_{\text{end}} + t_{\text{pad}})$$

If two padded intervals $[t'_{\text{start}}, t'_{\text{end}}]$ and $[t'_{\text{start}}, t'_{\text{end}}]$ satisfy:

$$t'_{\text{start}} + t_{\text{merge_threshold}} \leq t'_{\text{end}} \quad (t_{\text{merge_threshold}} = 100\text{ms})$$

They are merged into a single contiguous interval:

$$I_{\text{merged}} = [t'_{\text{start}}, \max(t'_{\text{end}}, t'_{\text{end}})]$$

3.2 Acoustic Energy Assertions

For every merged silencing interval I_k spanning sample range $[s_{\text{start}}, s_{\text{end}}]$ with $N_k = s_{\text{end}} - s_{\text{start}}$ samples:

$$\text{RMS}_k = \sqrt{\frac{1}{N_k} \sum_{n=s_{\text{start}}}^{s_{\text{end}}-1} x[n]^2}$$

$$\text{Peak}_k = \max_{n \in [s_{\text{start}}, s_{\text{end}}-1]} |x[n]|$$

Assertion Rule: In 'silence' mode, $\text{RMS}_k \leq 1 \times 10^{-5}$ and $\text{Peak}_k \leq 1 \times 10^{-5}$. Any higher residual energy triggers an immediate runtime exception and aborts the pipeline.

3.3 LINEAR16 WAV Encoding

Sample Rate: \$16,000\text{ Hz}\$

Channels: \$1\text{ (Mono)}\$

Bit Depth: \$16\text{-bit signed integer PCM (Little-Endian)}\$

Byte Rate: \$32,000\text{ bytes/sec}\$

Container: 44-byte standard RIFF WAVE header followed by 16-bit PCM payload.

4. Fail-Closed Verification Protocol (Constraint C8)

The redacted audio is passed directly to local ASR (simulated or WebGPU Whisper).

The resulting verification transcript is fed into IdentifierEngine.detectIdentifiers().

If any surviving identifier (structured, named entity, or special category) is found:

VolatileSessionStore.setVerificationFailure(survivors, verificationTranscript) is invoked.

isGatePassed is flipped to false.

The unredacted raw audio is preserved in volatile memory so the adviser can review and expand time boundaries.

Telemetry logs a redaction_verification_failed event without PII.

The UI directs the adviser back to Phase 9 with survivor cards.

If zero surviving identifiers are found:

LINEAR16 WAV buffer is created.

VolatileSessionStore.commitVerifiedRedactedAudio() overwrites rawAudioBuffer with zeros (Uint8Array.fill(0)) and sets the pointer to null.

Session advances to audio_redacted.